

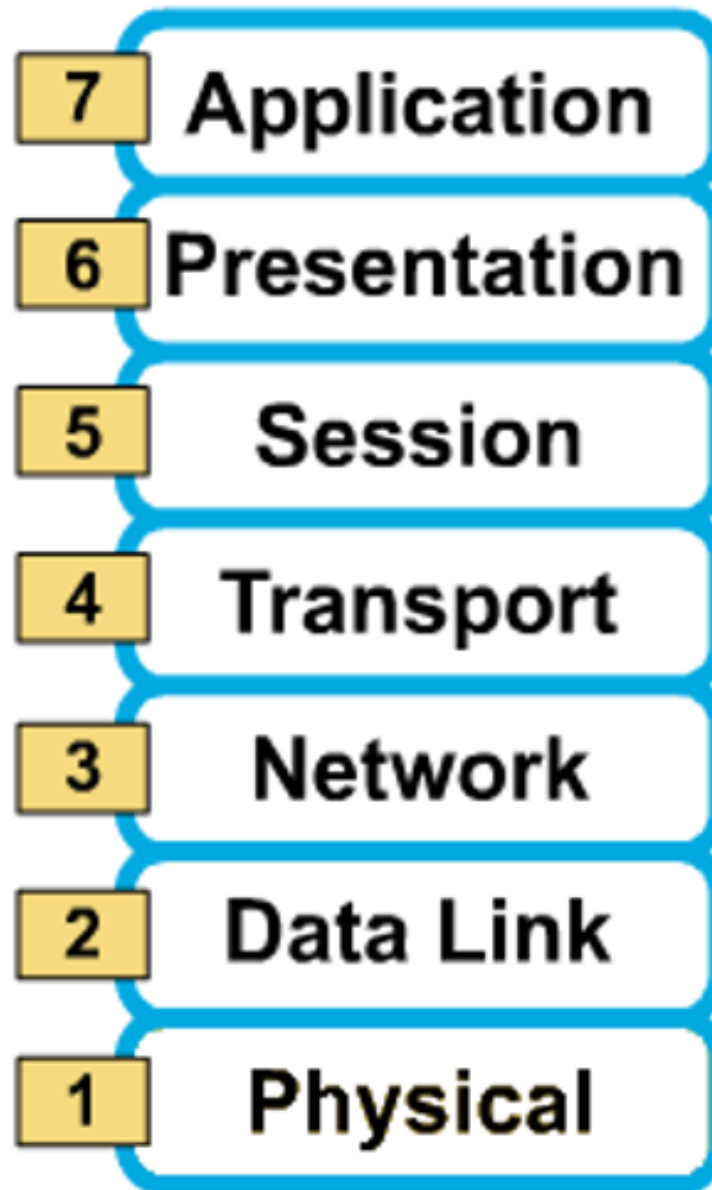
COMPUTER NETWORKS

Week-8

OSI

- The **Open System Interconnection Reference Model** (OSI Reference Model or **OSI Model**)
- An abstract description for layered communications and computer network protocol design
- → divides network architecture into seven layers

Seven Layers



How to remember?

- Please
 - Do
 - Not
 - Throw
 - Sausage
 - Pizza
 - Away
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- All People Seem To Need Data Processing
 - All Professors Should Teach Networking Daily Please

OSI

- In 1978, the International Standards Organization (ISO) began to develop its OSI framework architecture
- OSI has **two major components**:
- An abstract model of networking, called the Basic Reference Model or seven-layer model
- A set of specific protocols

Why Do We Need Layers?

- Break complex systems into manageable parts
- Easier troubleshooting
- Vendor interoperability

1. Application Layer

- Layer closest to the end user, which means that both the OSI application layer and the user interact directly with the software application
- Apps you use daily work at this layer
- Application layer **functions** typically include:
 - identifying communication partners
 - determining resource availability
 - synchronizing communication

ID comm. partners

- Before sending data, the system must identify the target device or service using:
- Domain names (e.g., google.com)
- IP addresses
- Application identifiers (like user accounts in apps)
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Determine Resource availability

- The application checks whether:
 - The destination device is online
 - The service (server) is available
 - Required resources (CPU, memory, bandwidth) are sufficient
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- On WhatsApp:
 - If your friend is offline → message shows single tick
 - If server is down → message won't send
- Calling someone → If their phone is off, you can't connect

Sync. Communication

- Ensures that both sender and receiver:
 - Follow the same rules/protocols
 - Exchange data in the correct order
 - Maintain proper time
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- While chatting on WhatsApp:
 - Messages appear in correct order
 - Typing indicators (“typing...”, “Recording...”) are synchronized
 - Read receipts (blue ticks) are updated correctly
 - Two people talking—if both speak at once or interrupt randomly, communication fails

Application Layer

- Some examples of application layer implementations include
 - Hypertext Transfer Protocol (HTTP)
 - File Transfer Protocol (FTP)
 - Simple Mail Transfer Protocol (SMTP)

2.Presentation Layer

- Establishes a context between Application Layer entities
- Responsible for encryption and decryption of data as well as compression and expansion of data
- It is sometimes called the syntax layer

Presentation Layer

- Translation
 - Encryption
 - Compression
 - Security: user authentication
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- Presentation layer defines the format in which the data is to be exchanged between the two communicating entities
 - Provides freedom from compatibility problems

3. Session layer

- Controls the dialogues (connections) between computers
- It establishes, manages and terminates the connections between the local and remote application
- It manages mode of communication (full-duplex, half-duplex, or simplex)
- Session control is managed (start, pause, resume, end)
- Manages recovery using checkpoints
- → manages sessions

Session Layer

- Simplex : Keyboard to computer, live TV broadcast, sensor sending data to a server
- No feedback possible
- Half Duplex: Classroom discussion, one side transmits at a time, avoids collision/confusion
- Full Duplex: Voice/Video calls on WhatsApp, Phone calls, Zoom meetings etc.
- Session layer ensures smooth, simultaneous exchange

Session Layer (Checkpointing)

- Inserts checkpoints during long data transfers
- To resume from the last checkpoint in case of failure instead of restarting everything
- E.g. for a 100MB file, checkpoints can be at 25MB, 50MB, 75MB
- In case a failure occurs at 60MB, resume from 50MB and not 0

Session Layer (Adjournment)

- Temporarily pause a session without ending it
- E.g. While uploading a file, Internet disconnects briefly
- Session is paused, not terminated

Session Layer (Termination)

- Cleanly end communication. E.g. Ending a call on WhatsApp, Logging out of a website
- This layer ensures that both sides agree communication is over and resources are released

Session Layer (Restart)

- Resume session after failure using checkpoints
- File download stops at 70%
- Resumes from last checkpoint
- Common in download managers and Cloud uploads (IDM: internet download manager)
- May not need it if you are working with small file downloads, stable internet and maybe onetime quick downloads
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4. Transport Layer

- The Transport Layer provides transparent transfer of data between end users, providing reliable data transfer services to the upper layers
- The Transport Layer controls the reliability of a given link through flow control, segmentation/desegmentation, and error control

Transport Layer

- The Transport Layer makes sure data is:
- Delivered correctly
- Delivered in order
- Delivered at the right speed

Transport Layer

- Carries out its functions through :
- Segmentation
- Desegmentation

Transport Layer

- Break big data into smaller pieces
- Large data is divided into segments
- Each segment is sent separately
- At the receiver → segments are reassembled (desegmentation)